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CODEBLOODED

Gamified Financial Visualizer

EPI-USE®

WHAT IS OUR GAMIFIED FINANCIAL VISUALIZER?

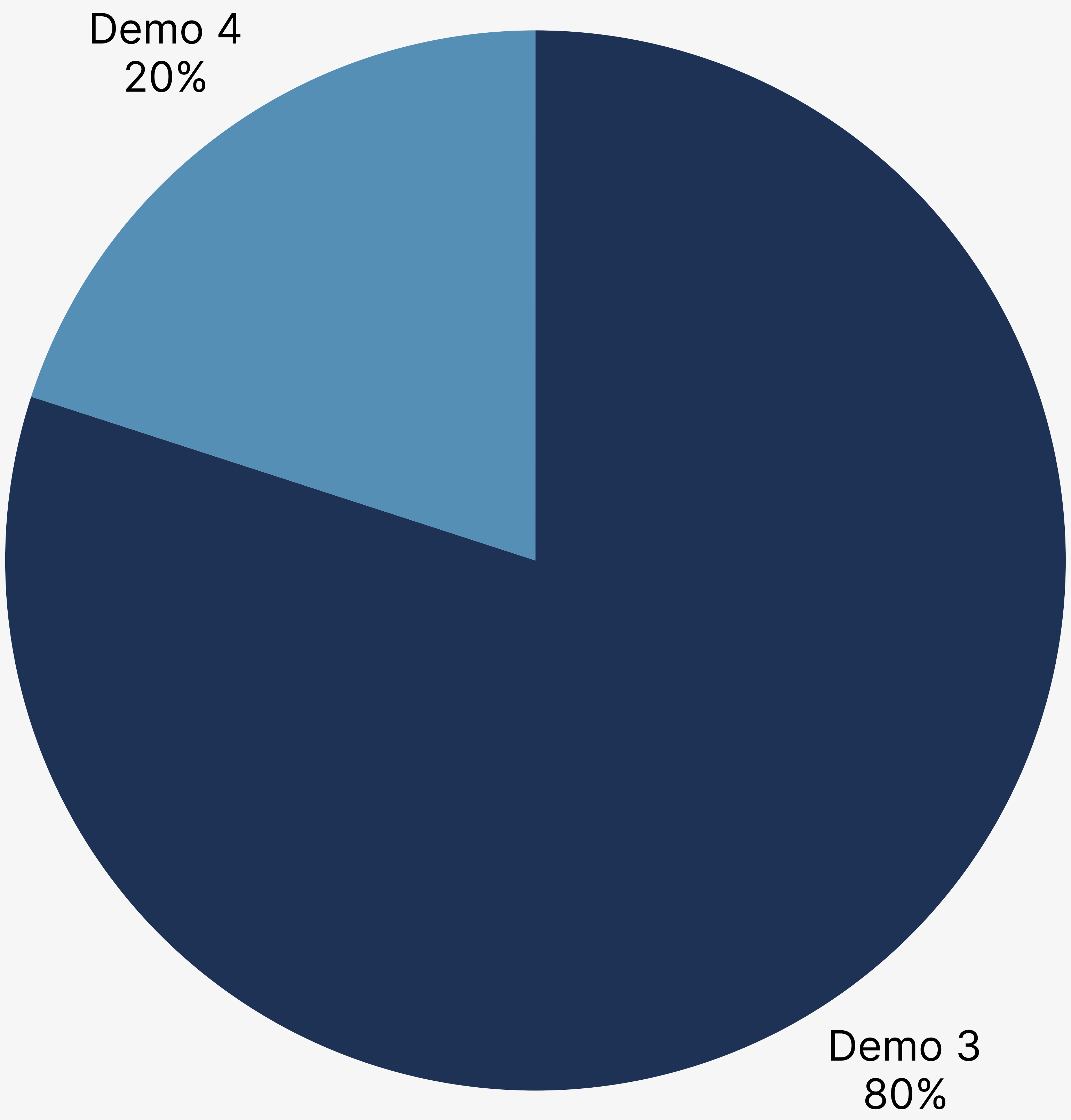
The **Gamified Financial Visualizer** is a cross-platform solution that leverages **AR** and **AI** to reimagine **financial literacy** as an engaging, game-like experience. Designed for digitally native users— particularly those interested in **financial management** —the platform makes **budgeting, saving, investing,** and **spending management** interactive and accessible. Aligned with **EPI-USE's** strategic goals, the solution integrates advanced **AI** and **Machine learning** to analyse transactions data and provide real-time, personalized financial advice contextualized to local realities.

Users can explore **financial goals** through immersive AR environments, engage with **AI financial coaches**, and navigate 3D "financial worlds." A built-in community layer promotes social accountability through **group challenges, shared milestones, and peer encouragement.** The platform supports **mobile** and **web access**, with future expansion planned for AR-enabled devices.

Since Demo 2

Demo distribution of the system

Completed system components vs incomplete components



System components: Demo 2 to 3

- AI Insights: Machine Learning
- Started AR configurations
- 3D Model
- AI Classifier
- UI Redesign
- Implemented most core requirements
  - Financial asset visualisation*
  - AI Expense tracking
  - Expense and income tracking
  - Community features
  - Comparative analysis
  - Goal setting
  - Educational content
  - AR Customisation*
- Implemented all optional requirements
- Small Usability Testing

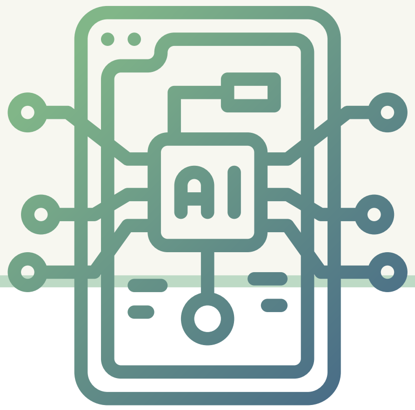
System components: Demo 4

- AI Integration (chat feature)
- Mobile (through PWA)
- Fully completed AR
- Community Game

SINCE DEMO 2

ML: Insights

- FastAPI-powered engine for analyzing spending, saving, and budgeting.
- Delivers rule-based summaries and charts with potential for ML expansion.
- Provides personalized insights to guide user financial behavior.

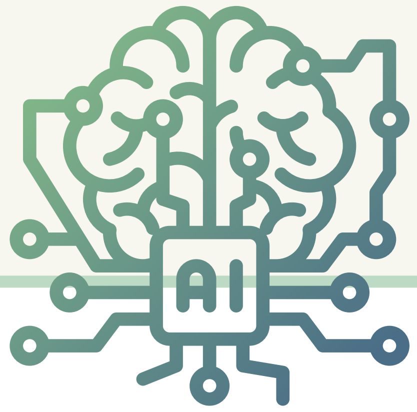


3D Model

- Developed a gamified city model using React Three Fiber + Blender.
- Recreated Blender's dual-sun lighting & shadow effects for realism.
- City reflects user progress across accounts, goals, and achievements.

AI: Classifier

- Automatically classifies transactions into categories (Food, Transport, etc.).
- Identifies income, expenses, and transfers accurately.
- Feeds into budgets, goals, and insights for consistency.



SINCE DEMO 2

Deployment

- PostgreSQL + Redis setup for data + caching in the cloud.
- Unified CI/CD pipeline with GitHub Actions for automated testing & deployment.

UI Update

- Full UI update with UX in mind.
- Community, accounts, profile, and learning page. expanded.
- Dashboard/home page revamp with the addition of a 3d model.
- Hotfixes to all pages.
- Additional explanations for the landing page.
- Added achievements page.

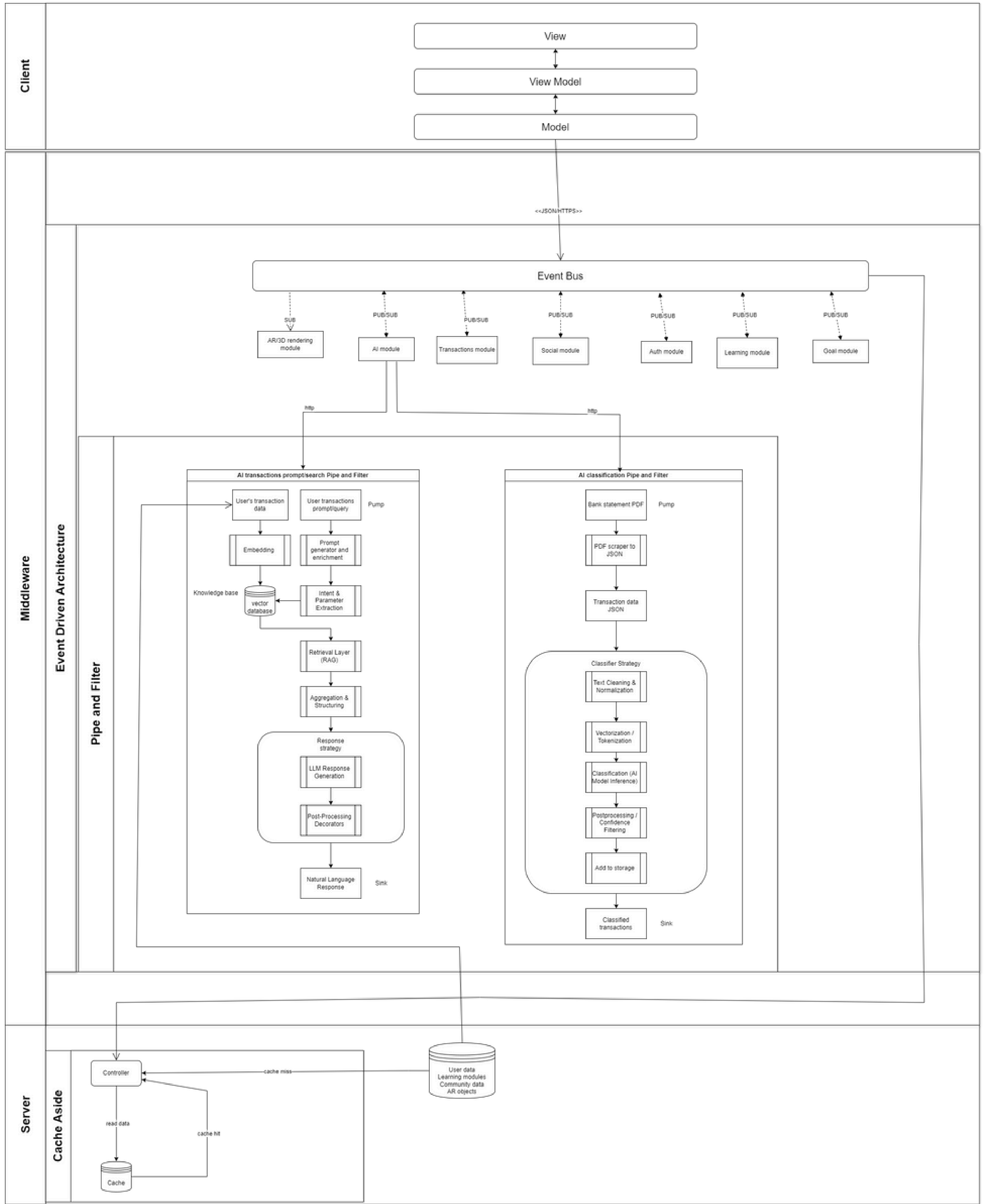
Architectural Implementation

- Modular, service-oriented system with clear separation (routes, services, DB).
- Frontend: React components for Accounts, Transactions, Goals, Insights, etc.
- Backend: Express + PostgreSQL with gamification features (XP, tiers, leaderboards).
- Designed for live syncing, scalability, and extensibility.

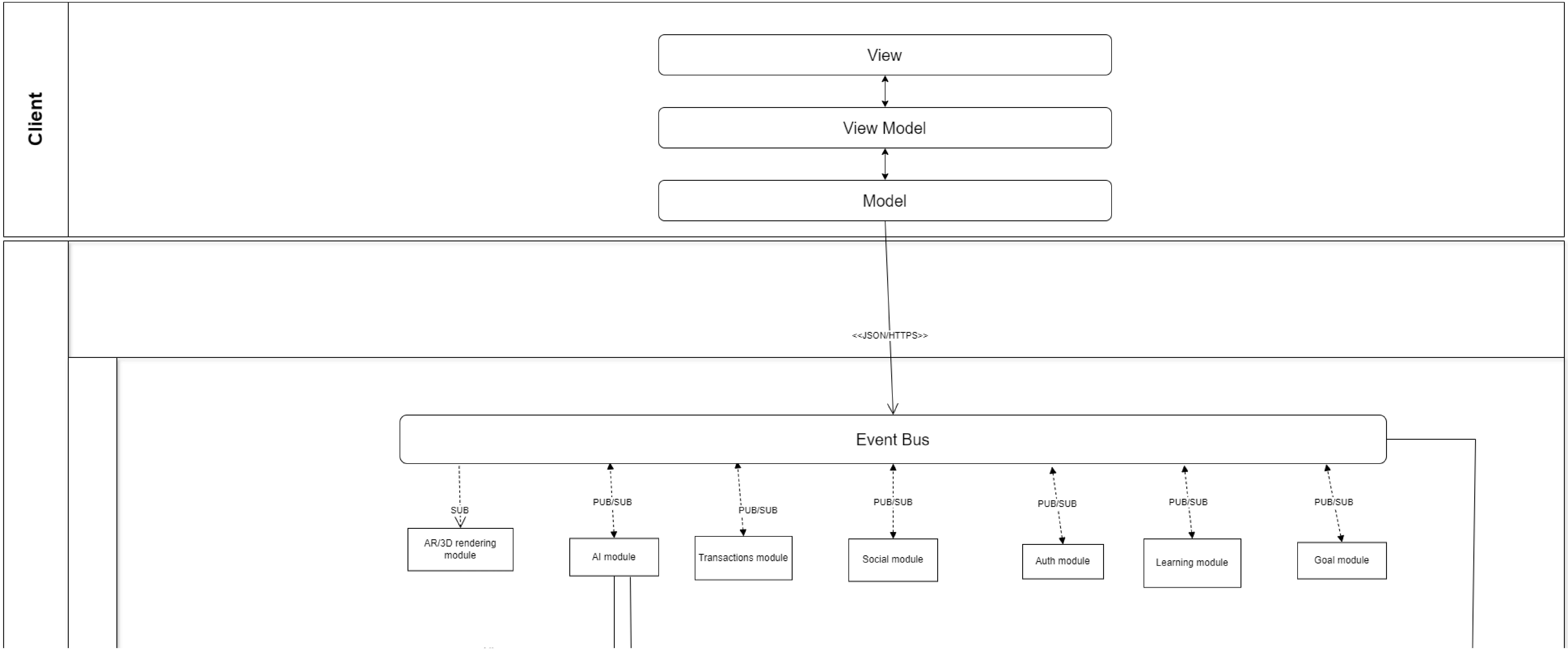
Live Demo

Architecture

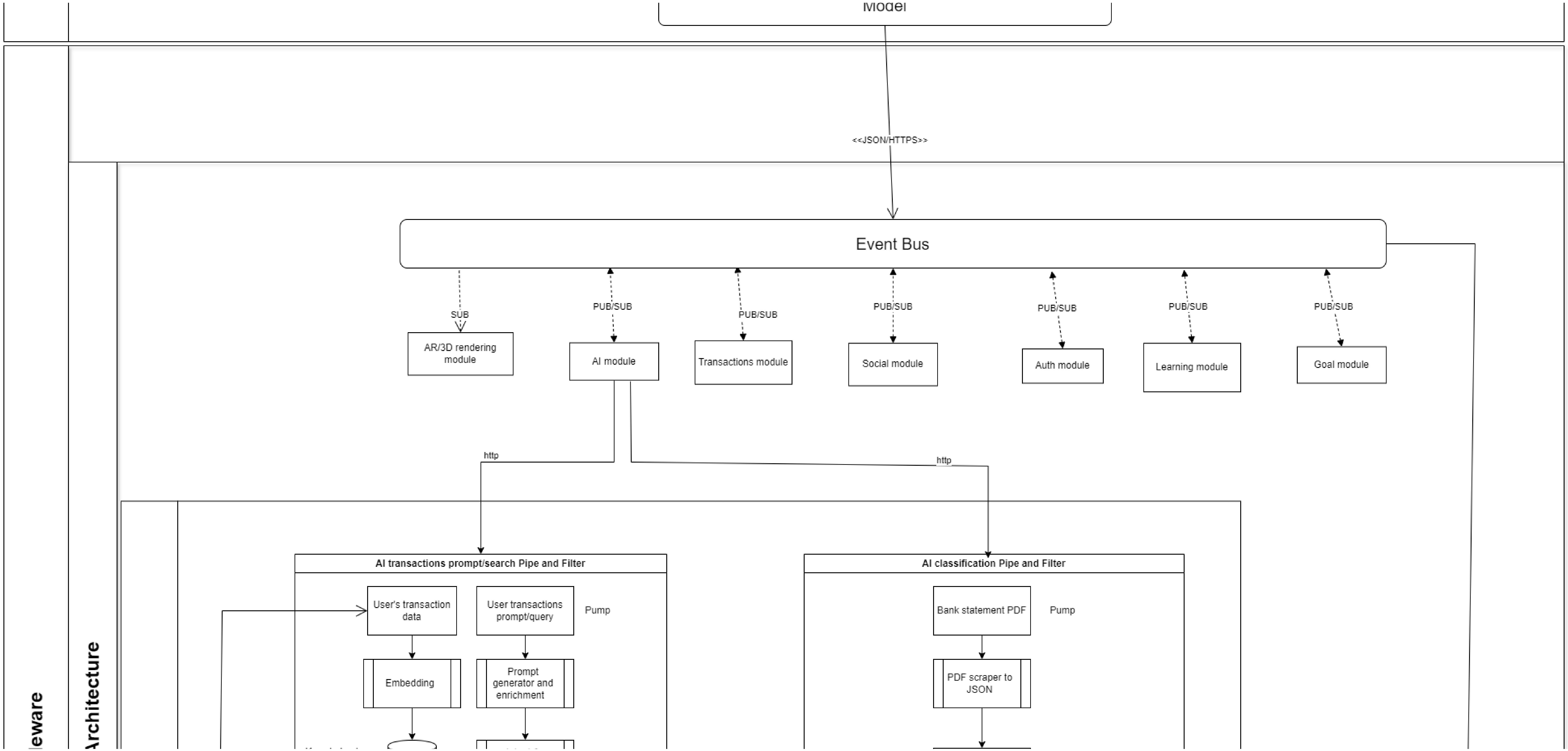
ARCHITECTURAL DIAGRAM



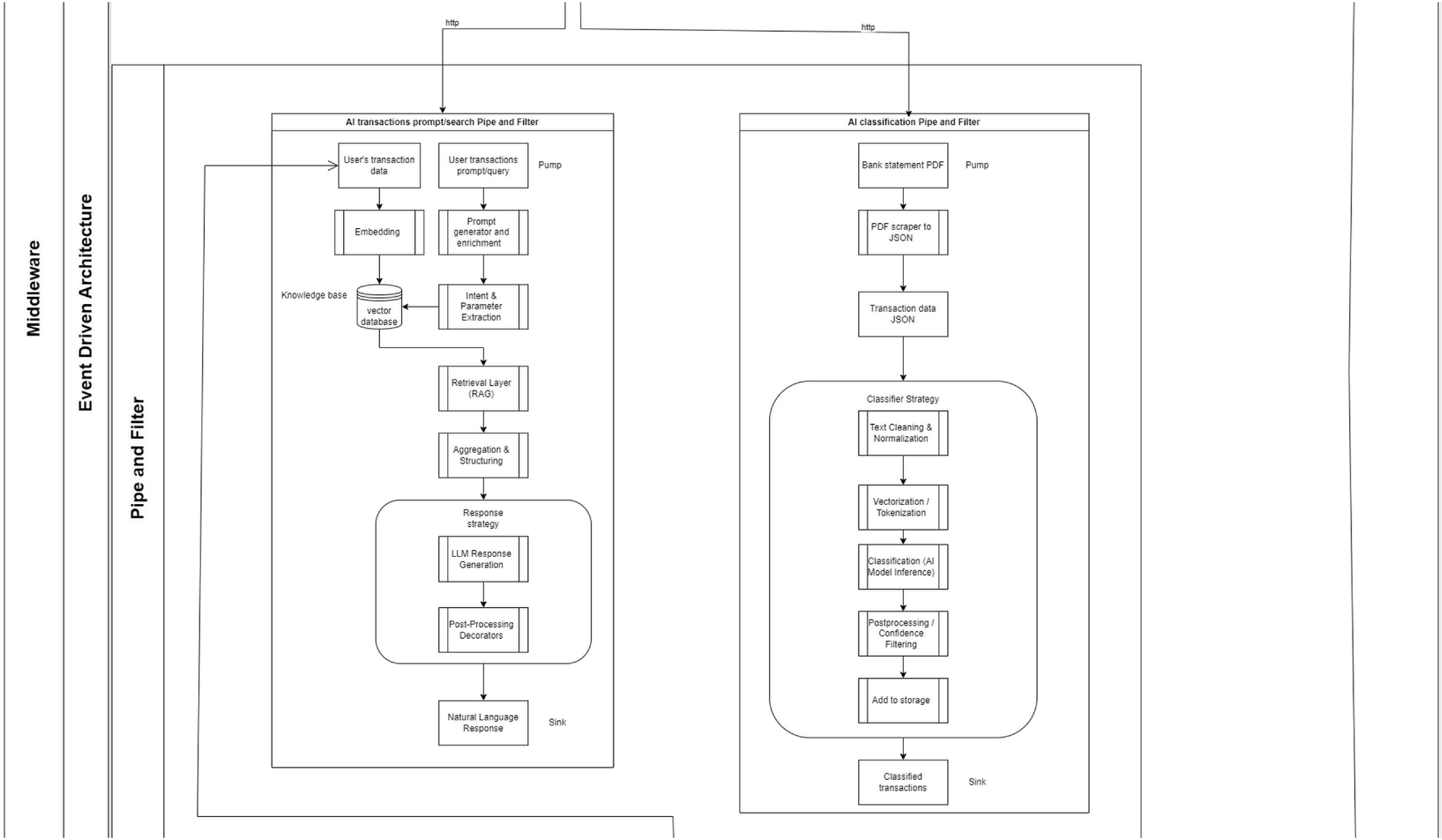
ARCHITECTURAL DIAGRAM



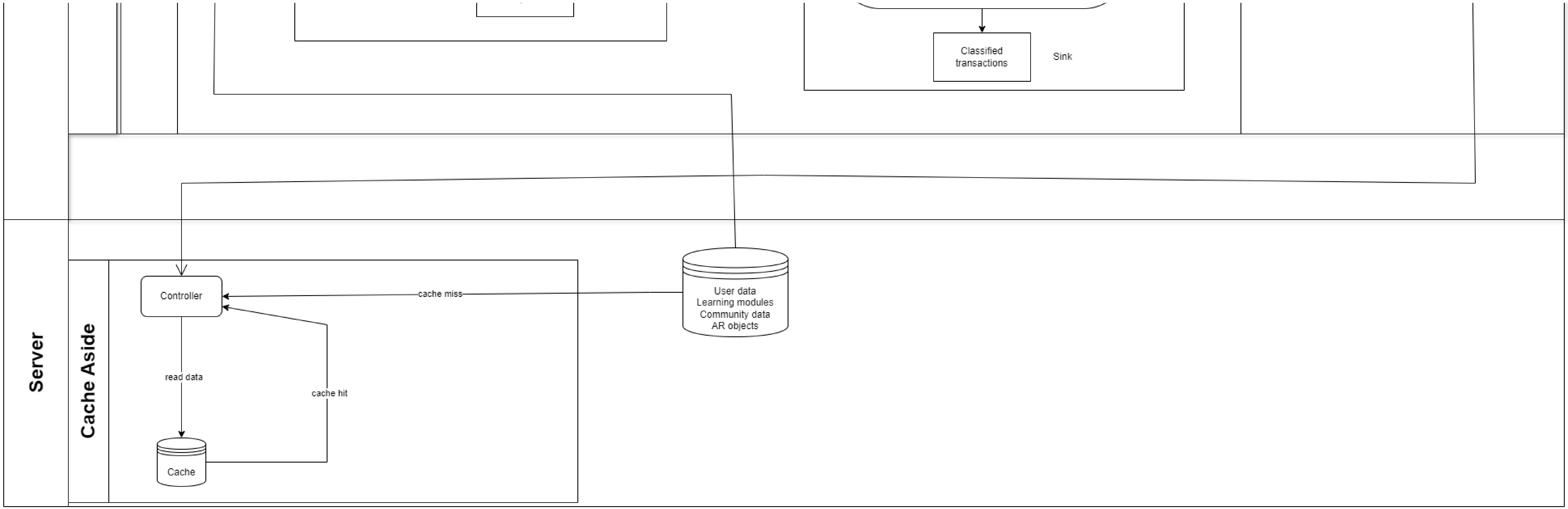
ARCHITECTURAL DIAGRAM



ARCHITECTURAL DIAGRAM



ARCHITECTURAL DIAGRAM



ARCHITECTURAL PATTERNS

Client-Server Architetcure

Modular subsystems within a unified client-server structure.
Each domain (Auth, Finance, AR, AI, etc.) has clear roles and interfaces.

Key Benefits:

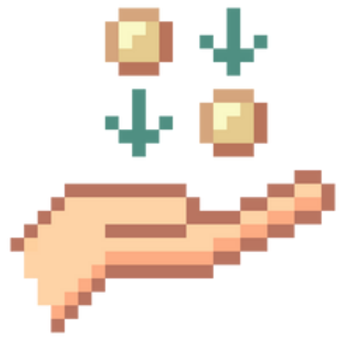
- Security – Controlled boundaries limit risks
- Scalability – Async workers handle heavy tasks
- Reliability – Subsystems fail independently
- Compatibility – Modules evolve without breaking others

Event-Driven Architecture

The system responds to real-time events like “goal achieved” or “quiz completed.”
Events are broadcast to multiple independent subscribers (e.g., notifications, AI modules).

Benefits (NFRs):

- Scalability – Add services without tight coupling
- Reliability – Non-blocking, fault-tolerant queues
- Usability – Instant user feedback
- Maintainability – Add logic without editing the core system



ARCHITECTURAL PATTERNS

Pipe and Filter

Processes data in sequential stages (filters) for AI pipelines and AR rendering. Each filter handles a specific task, like validation, transformation, or formatting.

Benefits (NFRs):

- Performance – Optimized per stage
- Reliability – Built-in validation/fallback
- Maintainability – Easy to test and extend
- Compliance – Insert legal filters (e.g., disclaimers)

Cache Aside

System checks cache (e.g., Redis) before querying the main DB.
Used for static or rarely changing data like goals or AR metadata.

Workflow:

- Check cache → return if hit
- If miss → query DB → cache → return

Benefits (NFRs):

- Performance – Faster access (<50ms)
- Scalability – Reduces DB pressure
- Reliability – Graceful degradation under load
- Maintainability – Transparent to service layers

ARCHITECTURAL PATTERNS

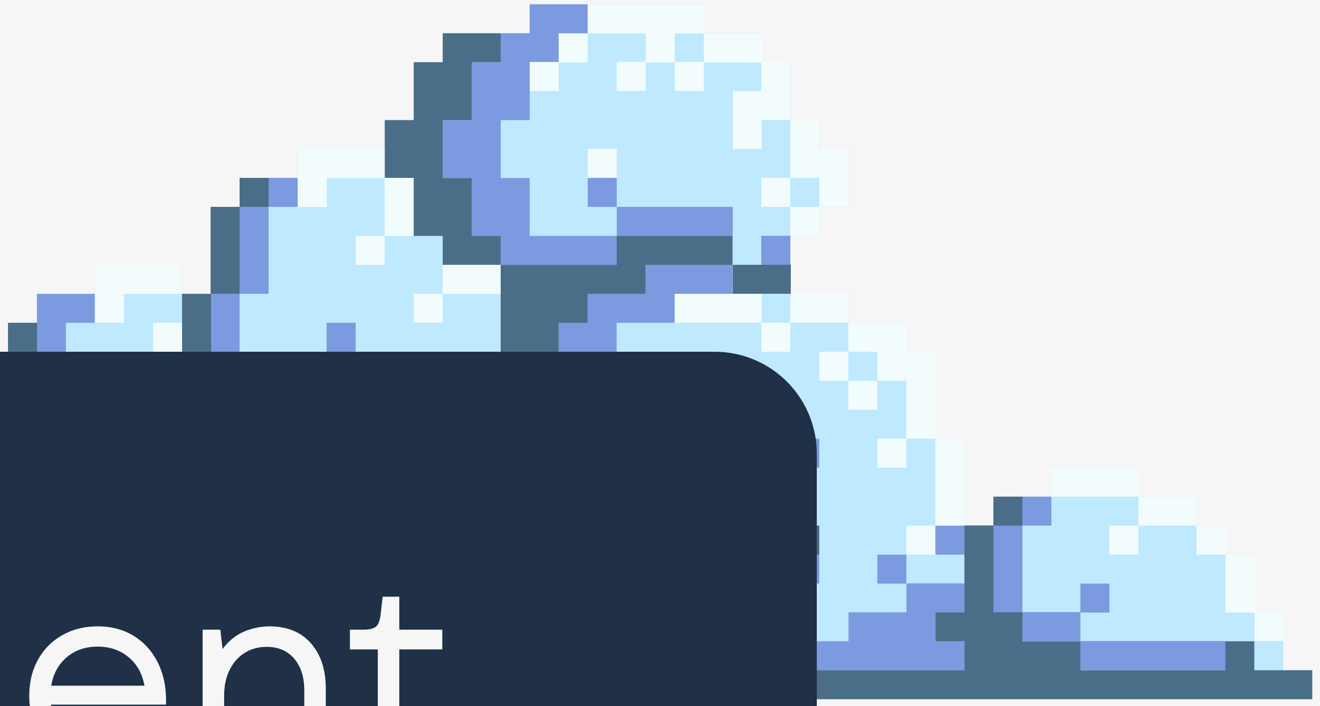
Model–View–ViewModel

Cache Aside

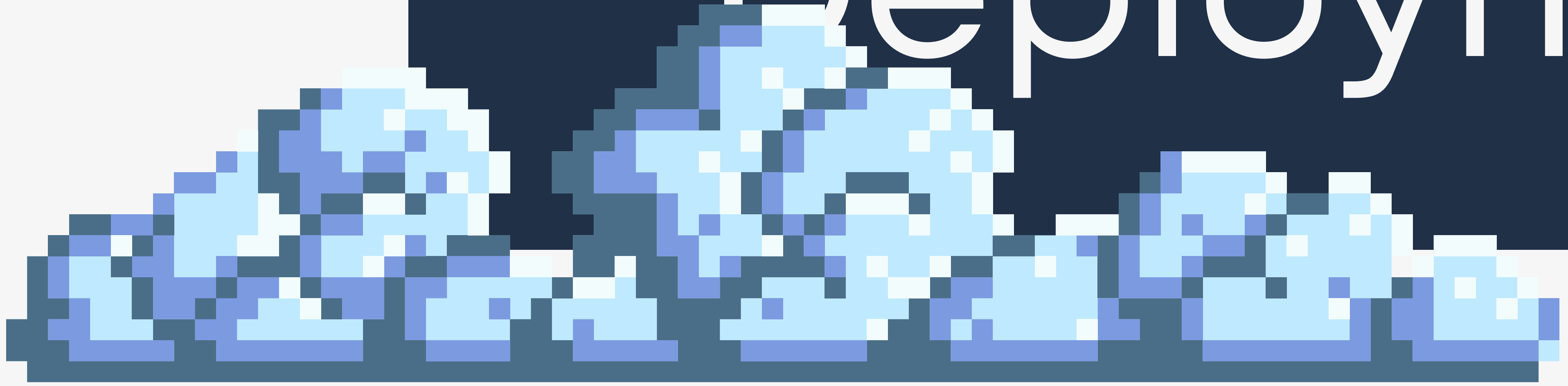
Processes UI logic in sequential layers for modular rendering and state synchronization. Each layer is responsible for a distinct concern, such as input handling, state transformation, or visual composition

Benefits (NFRs):

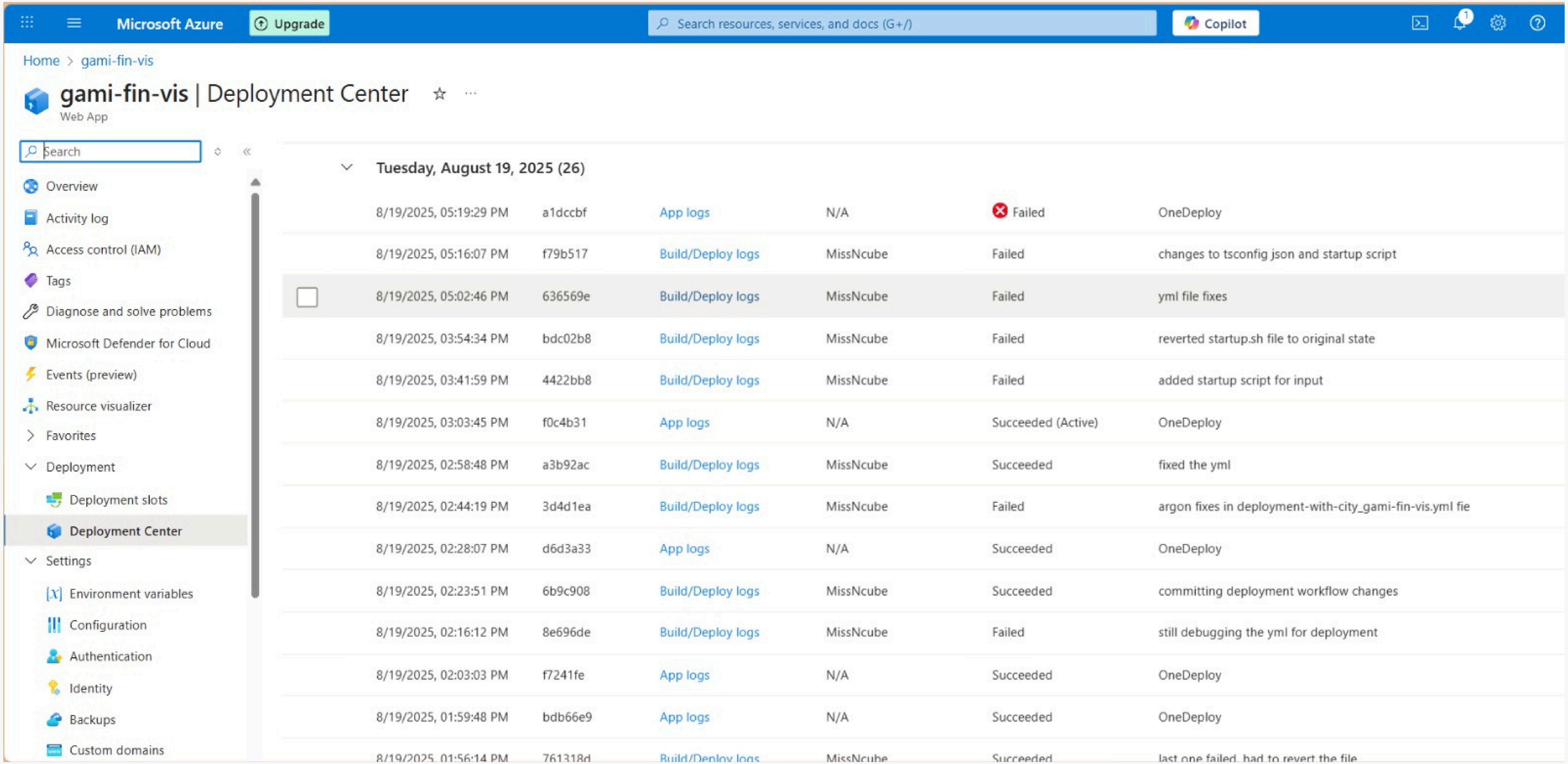
- Performance – Component-level re-renders reduce unnecessary DOM updates
- Reliability – Encapsulated logic and prop validation via TypeScript ensure predictable behavior
- Maintainability – Clear separation between views, hooks, and services makes it easy to test and refactor
- Scalability – New features can be added by composing new hooks or components without breaking existing ones
- Accessibility – Semantic HTML and ARIA tagging integrated at the presentation layer



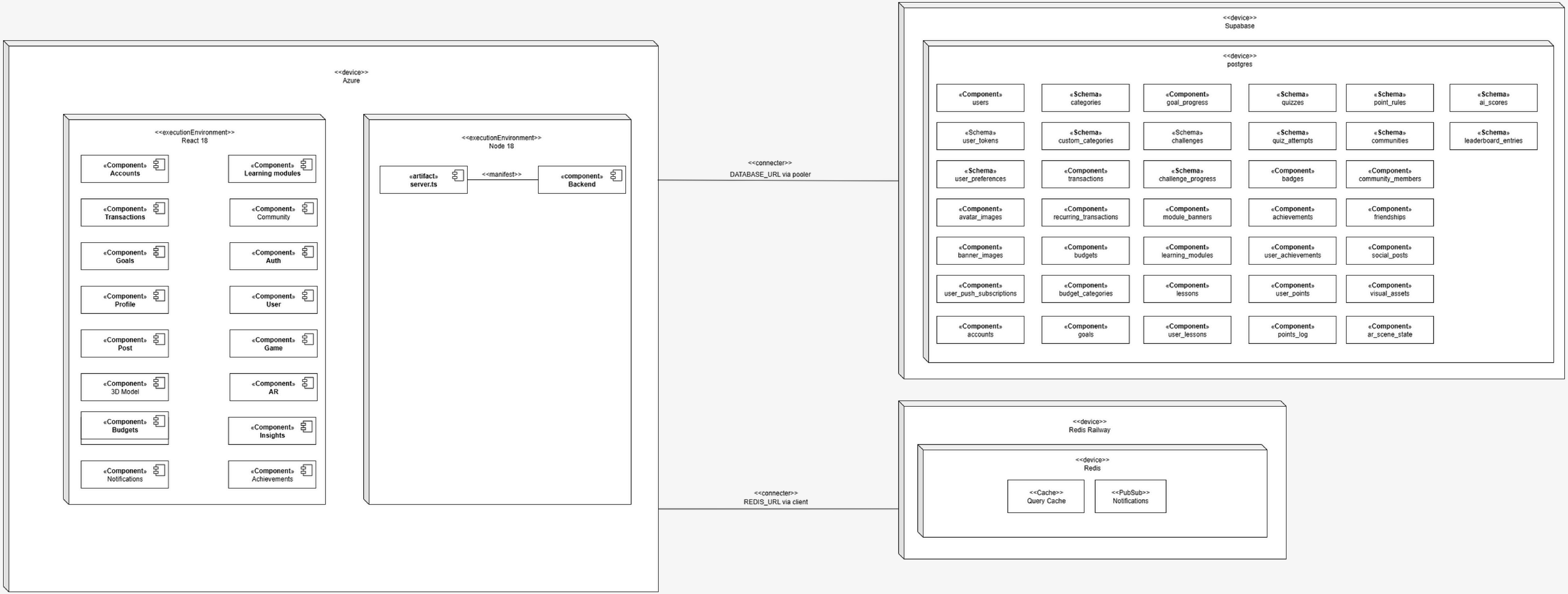
Deployment



DEPLOYMENT


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > gami-fin-vis
gami-fin-vis | Deployment Center
Web App
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Microsoft Defender for Cloud
Events (preview)
Resource visualizer
Favorites
Deployment
Deployment slots
Deployment Center
Settings
Environment variables
Configuration
Authentication
Identity
Backups
Custom domains
Tuesday, August 19, 2025 (26)
8/19/2025, 05:19:29 PM a1dccbf App logs N/A Failed OneDeploy
8/19/2025, 05:16:07 PM f79b517 Build/Deploy logs MissNcube Failed changes to tsconfig json and startup script
8/19/2025, 05:02:46 PM 636569e Build/Deploy logs MissNcube Failed yml file fixes
8/19/2025, 03:54:34 PM bdc02b8 Build/Deploy logs MissNcube Failed reverted startup.sh file to original state
8/19/2025, 03:41:59 PM 4422bb8 Build/Deploy logs MissNcube Failed added startup script for input
8/19/2025, 03:03:45 PM f0c4b31 App logs N/A Succeeded (Active) OneDeploy
8/19/2025, 02:58:48 PM a3b92ac Build/Deploy logs MissNcube Succeeded fixed the yml
8/19/2025, 02:44:19 PM 3d4d1ea Build/Deploy logs MissNcube Failed argon fixes in deployment-with-city_gami-fin-vis.yml fie
8/19/2025, 02:28:07 PM d6d3a33 App logs N/A Succeeded OneDeploy
8/19/2025, 02:23:51 PM 6b9c908 Build/Deploy logs MissNcube Succeeded committing deployment workflow changes
8/19/2025, 02:16:12 PM 8e696de Build/Deploy logs MissNcube Failed still debugging the yml for deployment
8/19/2025, 02:03:03 PM f7241fe App logs N/A Succeeded OneDeploy
8/19/2025, 01:59:48 PM bdb66e9 App logs N/A Succeeded OneDeploy
8/19/2025, 01:56:14 PM 761318d Build/Deploy logs MissNcube Succeeded last one failed, had to revert the file

DEPLOYMENT MODEL



Testing

UNIT TESTS

```
Test Suites: 7 passed, 7 total
Tests:       190 passed, 190 total
Snapshots:   0 total
Time:        7.505 s
Ran all test suites matching /services/i.
```

INTEGRATION TESTS

```
Test Suites: 7 passed, 7 total
Tests:       235 passed, 235 total
Snapshots:   0 total
Time:        857.475 s
Ran all test suites matching /integration/i.
```

INTEGRATION TESTS

```
================================================ test session starts =================================================
platform linux -- Python 3.10.12, pytest-8.4.1, pluggy-1.6.0
rootdir: /home/theplugza/Test Linux/Gamified-Financial-Visualizer/api-backend/modules/ai/app
plugins: anyio-4.9.0
collected 11 items

tests/insights_test.py ...........                                                                             [100%]

================================================= warnings summary ===================================================
tests/insights_test.py::test_category_shift
tests/insights_test.py::test_run_trend_analysis
  /home/theplugza/Test Linux/Gamified-Financial-Visualizer/api-backend/modules/ai/app/insights/services/trend_analysis.py:107: FutureWarning: DataFrameGrou
pBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the o
peration. Either pass `include_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.
    last_two.apply(lambda g: g.loc[g["amount"].idxmax(), "category"])

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
=========================================== 11 passed, 2 warnings in 2.16s ===========================================
```

CODE COVERAGE

Upload Jest coverage report 1s

```
▶ Run actions/upload-artifact@v4
With the provided path, there will be 48 files uploaded
Artifact name is valid!
Root directory input is valid!
Beginning upload of artifact content to blob storage
Uploaded bytes 300155
Finished uploading artifact content to blob storage!
SHA256 digest of uploaded artifact zip is d03dc0ebf008bef6c7524520aac19afa0fea0259c0170e0452672460741f3c99
Finalizing artifact upload
Artifact jest-coverage-report.zip successfully finalized. Artifact ID 3785853147
Artifact jest-coverage-report has been successfully uploaded! Final size is 300155 bytes. Artifact ID is 3785853147
Artifact download URL: https://github.com/COS301-SE-2025/Gamified-Financial-Visualizer/actions/runs/17033252114/artifacts/3785853147
```

COVERAGE REPORT

All files

78.73% Statements 1070/1359
64.35% Branches 186/289
92.36% Functions 121/131
79.15% Lines 1067/1348

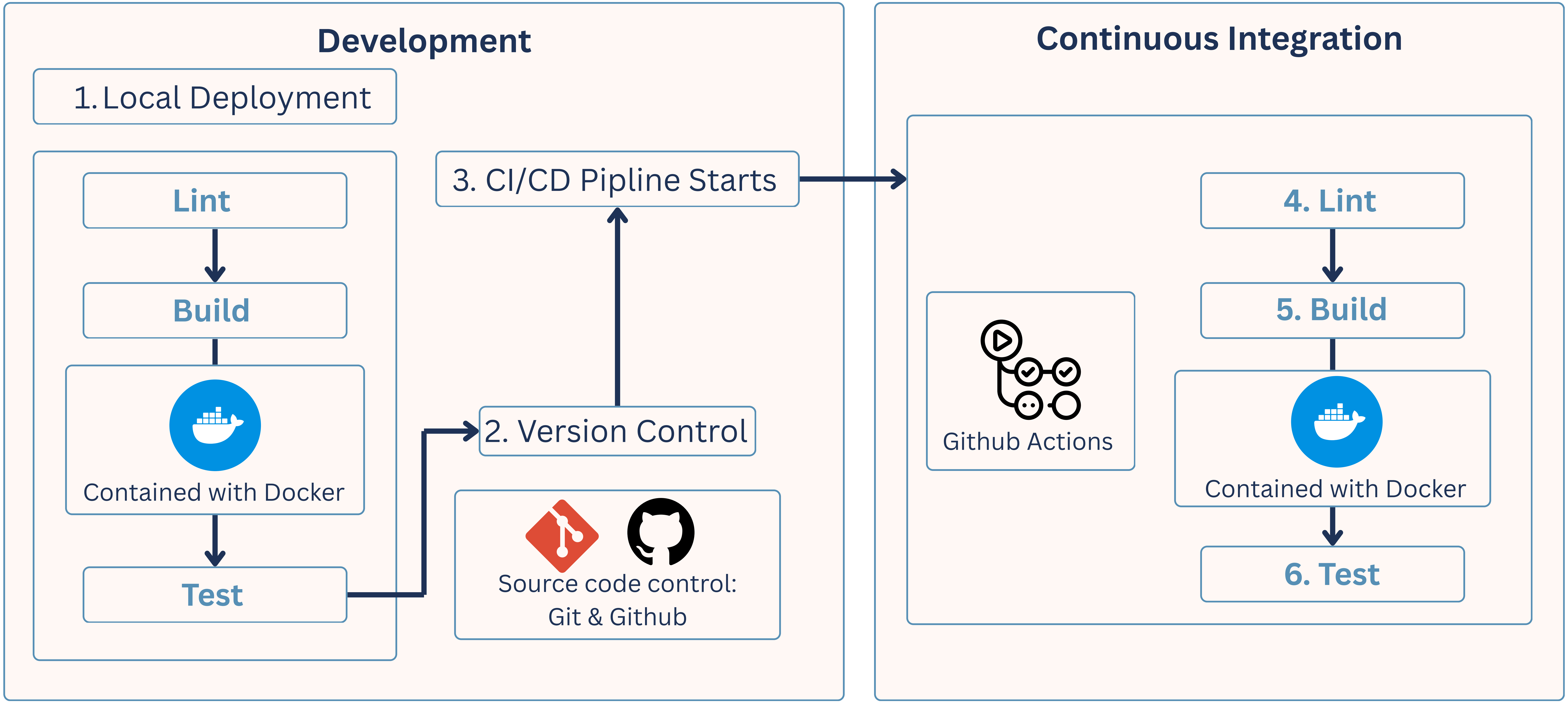
Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

File	Statements		Branches		Functions		Lines	
config	91.66%	11/12	70%	7/10	50%	1/2	100%	11/11
modules/auth/routes	97.59%	81/83	88.23%	15/17	100%	6/6	97.59%	81/83
modules/auth/services	75.13%	139/185	84%	21/25	85.71%	18/21	75.13%	139/185
modules/goals/routes	76.59%	108/141	63.15%	12/19	93.75%	15/16	76.59%	108/141
modules/goals/services	79.09%	140/177	57.69%	15/26	100%	16/16	78.85%	138/175
modules/learning/routes	56.98%	53/93	30%	3/10	80%	8/10	56.98%	53/93
modules/learning/services	89.92%	125/139	62.5%	25/40	100%	13/13	90.51%	124/137
modules/transactions/routes	79.14%	167/211	60.82%	59/97	85%	17/20	79.14%	167/211
modules/transactions/services	77.21%	244/316	64.44%	29/45	100%	26/26	78.7%	244/310
tests	100%	2/2	100%	0/0	100%	1/1	100%	2/2

Code coverage generated by istanbul at 2025-06-25T20:46:43.878Z

CI/CD

CI/CD PIPELINE



Documentation

DOCUMENTAION CHANGES

Requirements Specification

- Use case diagrams for implemented components
- Extension of functional requirements for implemented components

Technical Installation

- Prerequisites
- Installation
- Deployment/Running

Architectural Specification

- Quality Requirements
- Architectural Strategies
- Architectural Diagram
- Architectural Patterns
- Design Patterns
- Constraints
- Technology Requirements

User Manual

- Description of the project and instructions for use

ReadMe

- Brief project description
- Group member profiles
- Repo Badges
- Documenations and demo video

Coding Standards

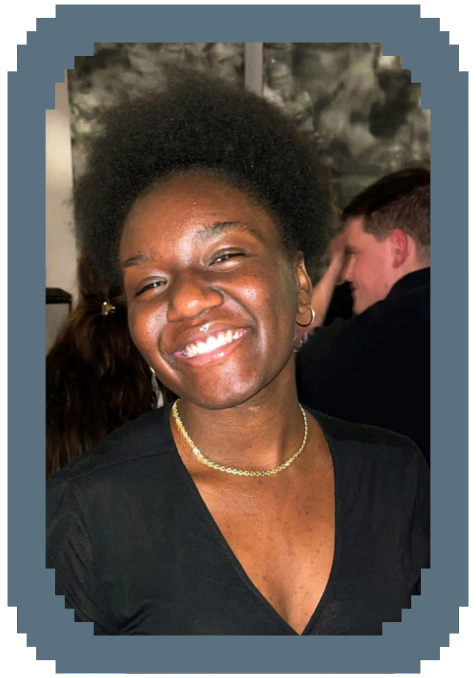
- Coding conventions and styles
- Branching strategy
- Linting strategy
- Guidelines

Service Contract

- API specifications of the backend services
- Definition of request and response bodies of routes

Contributions

INDIVIDUAL CONTRIBUTIONS



Yohali Malaika Kamangu

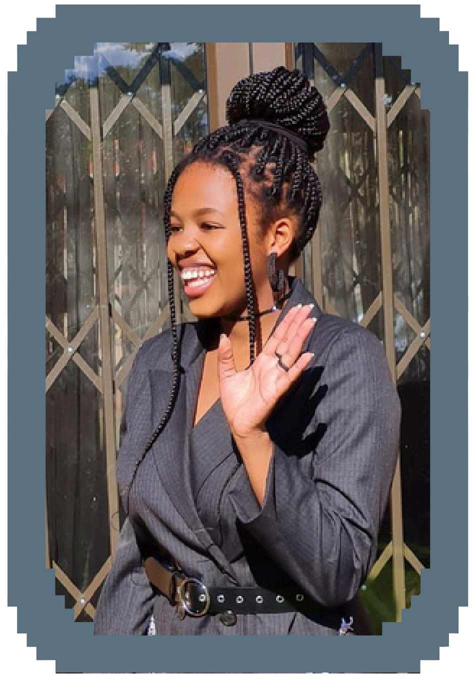
Project Manager,
Data Engineer

- Completed profile page
- Settings hotfixes
- Added/integrated the community posts feature
- Deployed database to Supabase
- Cleaned up database schema
- Seeded information
- Service contracts
- Database expansion

Ngaletsane Lebogang Masenya

Systems Architect,
Services Engineer

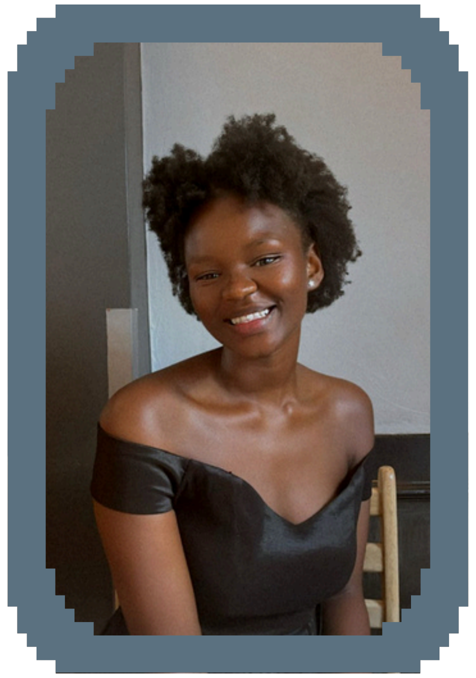
- Backen Update
- Architecture services
- AI categorization
- Machine learning insights services
- Technical Installation documentation
- Architectural specifics document update
- System Requirements document update
- Testing



Mpho Martha Siminya

UI/UX Designer,
UI Engineer

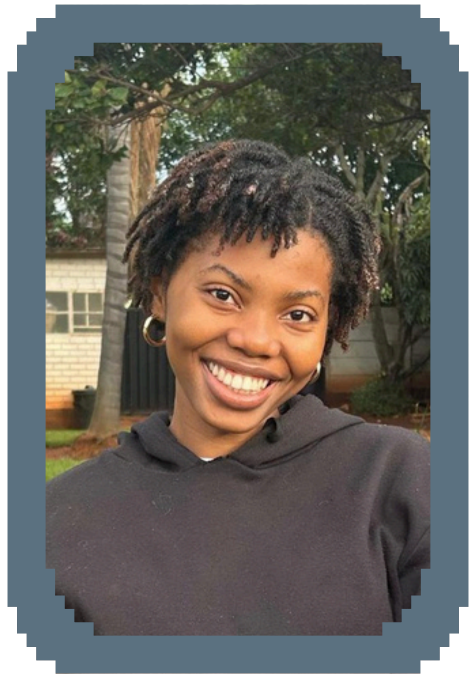
- Goals hotfixes
- Community hotfixes
- Achievements hotfixes
- Achievements banner design and creation
- 3D model design & creation
- Avatar design and creation
- UI redesign and update
- AR development
- Demo Presentation



Nobuhle Reitumetse Mtshali

DevOps, UI Engineer

- Completed dark mode.
- Accounts hotfixes
- Learn hotfixes
- Mobile creation
- Learn banner design and creation
- Profile hotfixes
- User manual documentation
- Dark mode integration
- DevOp improvements



Aundrea Nandie Ncube

Integration Engineer,
Testing Engineer

- Deployment model design and creation
- Backend services update and enhancement
- Security improvement
- Changed backend logic for the whole accounts page
- Integration of missing and new pages
- Team minutes

SRCUM FRAMEWORK

AGILE SCRUM

lightweight, iterative, team-driven framework for delivering working software incrementally.

Team Roles

- **Product Owner:** Bryan Janse van Vuuren and Elmar Jacobs
- **Scrum Master:** Malaika Kamangu (Project Mananger)
- **Development Team:** CodeBlooded Squad

Sprint Structure

Sprint Duration: Weekly

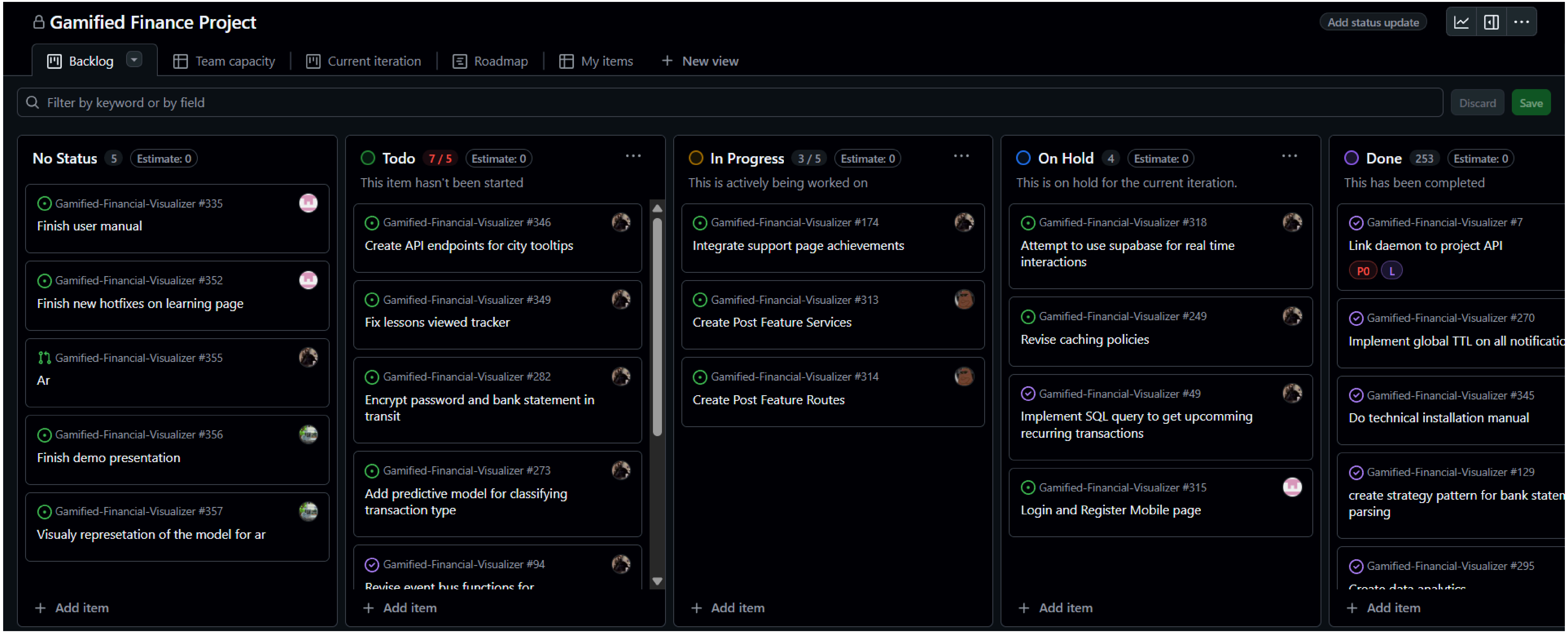
- Sprint Planning
- Daily Standups
- Sprint Review
- Sprint Retrospective

Tools & Practices

- **Projecting Tracking:** GitHub Projects/ Issues/ Milestones and Notion
- **Version Control:** Git (main, dev and feature branches)
- **Documentation:** README, ERDs, SRS and API specs
- **Continuous Integration:** GitHub Actions for testing workflows

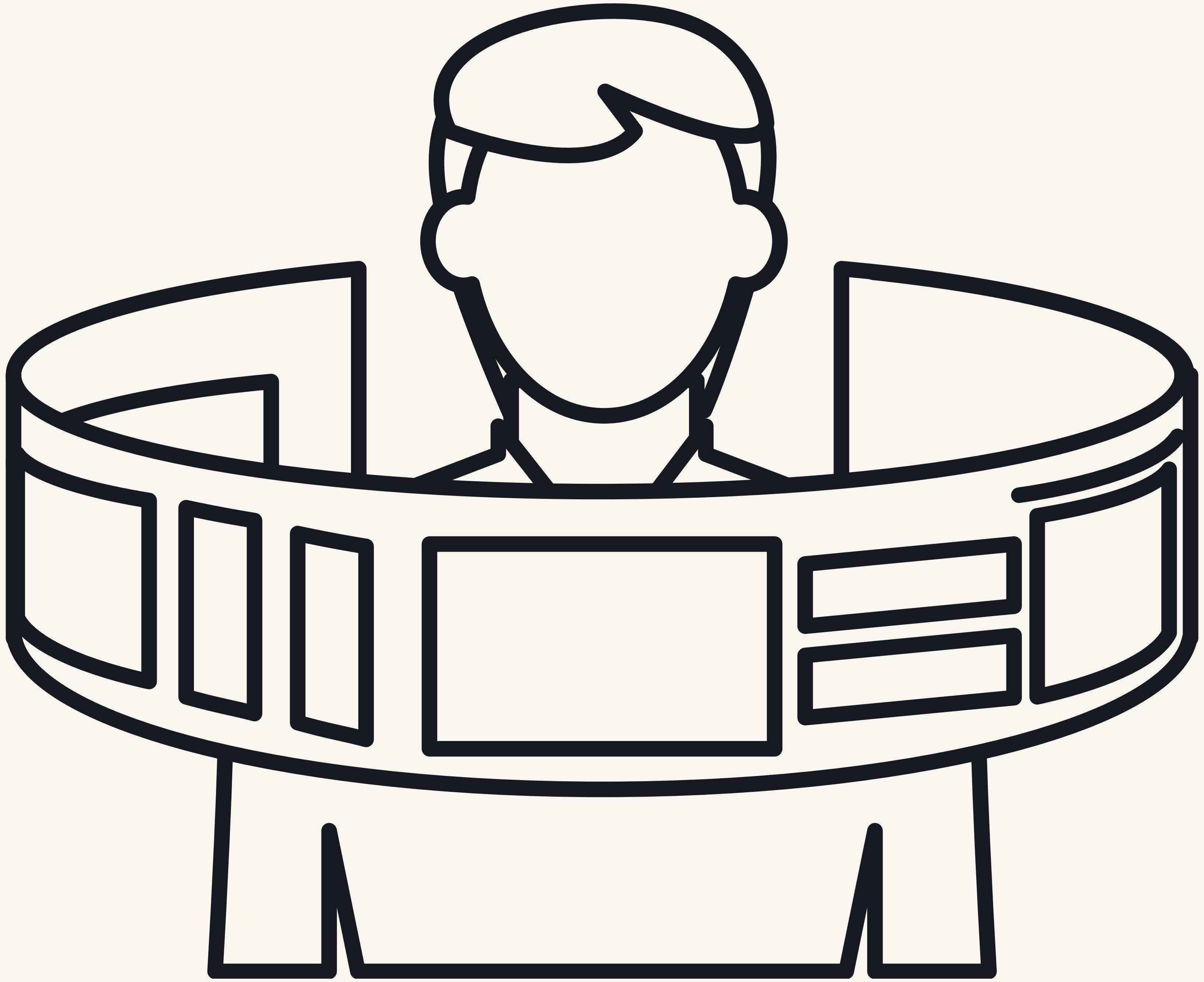
Notion Project Board

GROUP ORGANISATION



Demo 4

OUR FUTURE



Wow Factors

- Mobile
- 3D Model customization
- Community game
- AI Advisor

Testing & full deployment

- Usability Testing
- Performance Testing & Monitoring
- Full deployment of remaining features
- Security improvements

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Thank You.

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Capstone 2025